

Requirement Analysis - Solution Requirements

Parameter	Detail
Date	18 June 2026
Team ID	Book-a-Doctor Platform
Project Name	Book a Doctor
Maximum Marks	4 Marks

Functional Requirements

FR ID	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication & Authorization	- FR-1.1: Password encryption using Bcrypt. - FR-1.2: Dual-token authentication (Access Token + HttpOnly Cookies Refresh Token). - FR-1.3: Role-Based Access Control (RBAC) authorization middleware.
FR-2	Practitioner Vetting Queue	- FR-2.1: Doctor registration inputs (specialty, experience, consultation fee). - FR-2.2: Admin dashboard vetting view showing pending registrations. - FR-2.3: Admin toggle for doctor approval.
FR-3	Consultation Scheduling	- FR-3.1: Doctor slot scheduling calendar. - FR-3.2: Double-booking prevention check on active appointments. - FR-3.3: In-app notifications for appointment status changes (Pending, Approved, Rejected, Completed).
FR-4	Medical Document Locker	- FR-4.1: Multer upload parser for PDF/Image files. - FR-4.2: File restriction filter (limits size to 5MB, formats to PDF/JPG/PNG). - FR-4.3: Secure file service path mappings.

Non-functional Requirements

NFR ID	Non-Functional Requirement	Description	Target Metric / Technology
NFR-1	Usability	Intuitive user interface featuring light/dark mode and responsive layouts for mobile/desktop.	Bootstrap 5, glassmorphic styles, custom CSS variables
NFR-2	Security	Encryption of data in transit and at rest; protection of system routers against unauthorized roles.	Bcrypt (10 rounds), JWT auth guards, HttpOnly Cookies
NFR-3	Reliability	Integrity of booking operations; transactions must not allow clashing schedules.	Transactional checks in Mongoose, unit tests with Jest
NFR-4	Performance	High speed APIs responding to patient search queries and dashboard renders.	Database indices on doctor fields, query response < 200ms
NFR-5	Availability	System uptime and availability of doctors' schedules during consultations.	Uptime >= 99.9% (Render/MongoDB Atlas Cloud)
NFR-6	Scalability	Capacity to handle rising user volume and concurrent booking queries.	3-tier modular MVC design